

Ch 9 calculational

(c1)

(a) G-S on (f_1, f_2, f_3)

$$f_1 = e_1 + e_2 + e_3 = (1, 1, 1) \quad \|f_1\| = \sqrt{3}$$

$$f_2 = e_2 + e_3 = (0, 1, 1)$$

$$f_3 = e_3 = (0, 0, 1)$$

call the orthogonalized basis (F_1, F_2, F_3)

$$F_1 = \frac{f_1}{\|f_1\|} = \frac{e_1 + e_2 + e_3}{\|f_1\|} = \frac{1}{\sqrt{3}}(1, 1, 1)$$

$$A_2 = f_2 - \langle f_2, F_1 \rangle F_1 = f_2 - \frac{\langle f_2, f_1 \rangle}{\|f_1\|^2} f_1 \Rightarrow F_2 = \frac{A_2}{\|A_2\|} = \frac{(-2, 1, 1)}{\sqrt{6}}$$

$$= (0, 1, 1) - \frac{2}{3}(1, 1, 1) = \frac{1}{3}(-2, 1, 1)$$

$F_2 = \frac{(-2, 1, 1)}{\sqrt{6}}$

$$\approx (-2, 1, 1)$$

$$A_3 = f_3 - \langle f_3, F_2 \rangle F_2 - \langle f_3, F_1 \rangle F_1$$

$$= f_3 - \frac{\langle f_3, A_2 \rangle}{\|A_2\|^2} A_2 - \frac{\langle f_3, f_1 \rangle}{\|f_1\|^2} f_1 = (0, 0, 1) - \frac{1}{6}(-2, 1, 1) - \frac{1}{3}(1, 1, 1)$$

$$= \frac{1}{2}(0, -1, 1) \approx (0, -1, 1)$$

$\Rightarrow F_3 = \frac{(0, -1, 1)}{\sqrt{2}}$

Note the use of these intermediate orthogonal vectors

A_j where $F_j = \frac{A_j}{\|A_j\|}$.

we can ignore constant factors with them because we only care about their direction, not magnitude.

(b) G-S on (f_3, f_2, f_1)

$$F_1 = \frac{f_3}{\|f_3\|} = f_3 = e_3$$

$$F_2 = f_2 - \langle f_2, F_1 \rangle F_1$$

$$= f_2 - \langle f_2, f_3 \rangle f_3 = f_2 - f_3 = (0, 1, 0) = e_2$$

$$F_3 = f_1 - \langle f_1, F_2 \rangle F_2 - \langle f_1, F_1 \rangle F_1$$

$$= f_1 - \langle f_1, e_2 \rangle e_2 - \langle f_1, e_3 \rangle e_3$$

$$= (1, 1, 1) - (0, 1, 0) - (0, 0, 1) = (1, 0, 0) = e_1$$

Thus $(F_1, F_2, F_3) = (e_1, e_2, e_3)$

② Let $\mathcal{F} = \left\{ \frac{1}{\sqrt{2\pi}}, \frac{\sin kx}{\sqrt{\pi}}, \frac{\cos kx}{\sqrt{\pi}} \mid k=1,2,\dots,n \right\} \subset [-\pi, \pi]$

Normal we show that $\|f\|=1$

for $f \in \mathcal{F}$, since $\|f\|^2 = \langle f, f \rangle = \int_{-\pi}^{\pi} f^2(x) dx$

• For $f = \frac{1}{\sqrt{2\pi}}$, $\|f\|^2 = \int_{-\pi}^{\pi} \frac{1}{2\pi} dx = \frac{2\pi}{2\pi} = 1$

• $f = \frac{\sin kx}{\sqrt{\pi}} \Rightarrow \|f\|^2 = \int_{-\pi}^{\pi} \sin^2 kx dx = \frac{1}{2\pi} \int_{-\pi}^{\pi} (1 - \cos 2kx) dx$
 $= \frac{1}{2\pi} \left[2\pi - \frac{1}{2k} \sin 2kx \right]_{-\pi}^{\pi} = 1 - \frac{1}{2k} (\sin 2k\pi + \sin 2k\pi) = 1$

• $f = \frac{\cos kx}{\sqrt{\pi}} \Rightarrow \|f\|^2 = \int_{-\pi}^{\pi} \cos^2 kx dx = \int_{-\pi}^{\pi} \frac{1 + \cos 2kx}{2} dx = 1$

for any $k \in \mathbb{N}$, Thus every $f \in \mathcal{F}$ is a unit vector.

Orthogonal we show that for any $f_i, f_j \in \mathcal{F}$, $f_i \neq f_j$

$\langle f_i, f_j \rangle = \int_{-\pi}^{\pi} f_i(x) f_j(x) dx = 0$

* Note that constants don't affect orthogonality,

i.e. $\langle f, g \rangle = 0 \Rightarrow \langle \alpha f, \beta g \rangle = 0$ for any $\alpha, \beta \in \mathbb{R}$

For $k, l \in \mathbb{Z}$, $k \neq l$,

• $\langle \sin kx, \sin lx \rangle = \int_{-\pi}^{\pi} \sin kx \sin lx dx$
 $= \frac{1}{2} \int_{-\pi}^{\pi} (\cos(k-l)x - \cos(k+l)x) dx$
 $= \frac{1}{2} \left[\frac{\sin(k-l)x}{k-l} - \frac{\sin(k+l)x}{k+l} \right]_{-\pi}^{\pi} = 0$

because $\sin m\pi = 0$ for any $m \in \mathbb{Z}$.

• $\langle \cos kx, \cos lx \rangle = \int_{-\pi}^{\pi} \cos kx \cos lx dx$
 $= \frac{1}{2} \int_{-\pi}^{\pi} [\cos(k+l)x + \cos(k-l)x] dx = 0$

for any $k, l \in \mathbb{Z}$. ($k=l$ is ok here),

• $\langle \cos kx, \sin lx \rangle = \int_{-\pi}^{\pi} \cos kx \sin lx dx = 0$

because $\cos kx$ is an even function
 $\sin lx$ is an odd function

Then $\cos kx \sin lx$ is an odd function.

Hence, $\mathcal{F}$ is an orthonormal set of vectors.

③ Ortho normalize $\{1, x, x^2\} \subset \mathbb{R}_2[x]$ with respect to the inner product $\langle f, g \rangle = \int_0^1 f(x)g(x) dx$ for $f, g \in \mathbb{R}_2[x]$

$\|1\|^2 = \langle 1, 1 \rangle = \int_0^1 dx = 1 \Rightarrow f_1(x) = \frac{1}{\|1\|} = 1$

$A_2 = x - \langle x, f_1 \rangle f_1 = x - \int_0^1 x dx = x - \frac{1}{2}$
 $\|A_2\|^2 = \int_0^1 (x - \frac{1}{2})^2 dx = \frac{1}{3} (x - \frac{1}{2})^3 \Big|_0^1 = \frac{1}{12}$
 $\Rightarrow f_2 = \frac{A_2}{\|A_2\|} = \sqrt{12} \left(x - \frac{1}{2} \right)$

$A_3 = x^2 - \langle x^2, f_2 \rangle f_2 - \langle x^2, f_1 \rangle f_1 = x^2 - \frac{\langle x^2, A_2 \rangle}{\|A_2\|^2} A_2 - \langle x^2, f_1 \rangle f_1$

$\langle x^2, A_2 \rangle = \int_0^1 x^2 (x - \frac{1}{2}) dx = \frac{1}{12}$

$\langle x^2, f_1 \rangle = \int_0^1 x^2 dx = \frac{1}{3}$

$A_3 = x^2 - \frac{12}{12} (x - \frac{1}{2}) - \frac{1}{3} = x^2 - x + \frac{1}{6}$
 $\Rightarrow f_3 = \sqrt{180} \left(x^2 - x + \frac{1}{6} \right)$

$\|A_3\|^2 = \int_0^1 (x^2 - x + \frac{1}{6})^2 dx = \frac{1}{180}$

④ Orthonormalize $\{v_1, v_2, v_3\} \subset \mathbb{R}^3$ w.r.t. standard inner product on $\mathbb{R}^3$.

$$v_1 = (1, 2, 1), \quad v_2 = (1, -2, 1), \quad v_3 = (1, 2, -1)$$

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{6}}(1, 2, 1)$$

$$v_2 - \langle v_2, u_1 \rangle u_1 = (1, -2, 1) - \frac{-2}{6}(1, 2, 1) = \frac{4}{3}(1, -1, 1)$$

Let $A_2 = (1, -1, 1)$. Note that $A_2 \perp u_1$.

$$u_2 = \frac{A_2}{\|A_2\|} = \frac{1}{\sqrt{3}}(1, -1, 1)$$

$$v_3 - \langle v_3, u_2 \rangle u_2 - \langle v_3, u_1 \rangle u_1 = v_3 - \frac{\langle v_3, A_2 \rangle A_2}{\|A_2\|^2} - \frac{\langle v_3, v_1 \rangle}{\|v_1\|^2} v_1$$

$$= (1, 2, -1) - \frac{-2}{3}(1, -1, 1) - \frac{4}{6}(1, 2, 1)$$

$$= (1, 0, -1)$$

$$u_3 = \frac{1}{\sqrt{2}}(1, 0, -1)$$

Math 67 – HW 7 Proof Solutions

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Chapter 7

P1. Let V be a finite dimensional inner product space over $\mathbb{F}$. For any $u, v \in V$ the following are equivalent.

- (a) $\langle u, v \rangle = 0$
- (b) $\|u\| \leq \|u + \alpha v\|$ for every $\alpha \in \mathbb{F}$.

Proof. This is trivial for $v = 0$, so we will assume that $v \neq 0$.

(a) $\Rightarrow$ (b): Assume $\langle u, v \rangle = 0$. Note that scaling of vectors does not affect orthogonality; i.e. $\langle u, v \rangle = \langle \beta u, \gamma v \rangle = \langle \beta u, \gamma v \rangle = 0$ for any scalars $\beta, \gamma \in \mathbb{F}$. Then

$$\begin{aligned}\|u + \alpha v\|^2 &= \langle u + \alpha v, u + \alpha v \rangle \\ &= \langle u, u \rangle + \langle u, \alpha v \rangle + \langle \alpha v, u \rangle + \langle \alpha v, \alpha v \rangle \\ &= \|u\|^2 + \|\alpha v\|^2.\end{aligned}$$

Since $\|\alpha v\|^2 \geq 0$ it follows that $\|u + \alpha v\|^2 \geq \|u\|^2$, and thus $\|u + \alpha v\| \geq \|u\|$.

(b) $\Rightarrow$ (a): We assume $\|u + \alpha v\|^2 \geq \|u\|^2$ for any $\alpha \in \mathbb{F}$. Then

$$\begin{aligned}0 &\leq \|u + \alpha v\|^2 - \|u\|^2 \\ &= \langle u, \alpha v \rangle + \langle \alpha v, u \rangle + \langle \alpha v, \alpha v \rangle \\ &= \overline{\alpha} \langle u, v \rangle + \alpha \overline{\langle u, v \rangle} + |\alpha|^2 \|v\|^2.\end{aligned}$$

Since this holds for any $\alpha \in \mathbb{F}$, and $v \neq 0$, let $\alpha = -\langle u, v \rangle / \|v\|^2$. Then we have

$$\begin{aligned}0 &\leq -\frac{\overline{\langle u, v \rangle} \langle u, v \rangle}{\|v\|^2} - \frac{\langle u, v \rangle \overline{\langle u, v \rangle}}{\|v\|^2} + \left| \frac{\langle u, v \rangle}{\|v\|^2} \right|^2 \|v\|^2 \\ &= -2 \frac{|\langle u, v \rangle|^2}{\|v\|^2} + \frac{|\langle u, v \rangle|^2}{\|v\|^2} \\ &= -\frac{|\langle u, v \rangle|^2}{\|v\|^2}.\end{aligned}$$

Then $|\langle u, v \rangle|^2 \leq 0$, which is only possible if $\langle u, v \rangle = 0$. $\square$

Note: The choice of $\alpha = -\langle u, v \rangle / \|v\|^2$ may not seem very intuitive, but if you read over page 121 in the text (after the proof of Pythagorean Theorem) it might make sense.

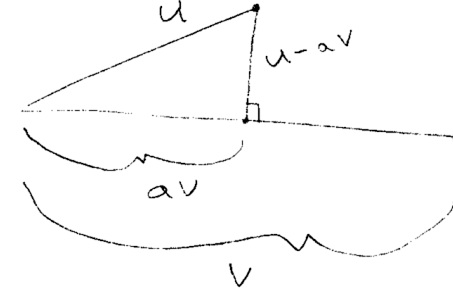


Figure 1: Here, $a = \langle u, v \rangle / \|v\|^2$ so that $u - av$ is the component of u which is orthogonal to v .

P2. Let $n \in \mathbb{Z}_+$, and let $a_1, \dots, a_n, b_1, \dots, b_n \in \mathbb{R}$. Then

$$\left(\sum_{k=1}^n a_k b_k \right)^2 \leq \left(\sum_{k=1}^n k a_k^2 \right) \left(\sum_{k=1}^n \frac{b_k^2}{k} \right)$$

Proof. This follows directly from the Cauchy-Schwarz inequality (Theorem 9.3.3)

$$\begin{aligned}|\langle u, v \rangle| &\leq \|u\| \|v\| \\ |\langle u, v \rangle|^2 &\leq \|u\|^2 \|v\|^2 \\ \left(\sum_{k=1}^n u_k v_k \right)^2 &\leq \left(\sum_{k=1}^n u_k^2 \right) \left(\sum_{k=1}^n v_k^2 \right)\end{aligned}$$

if we let $u = (a_1, a_2\sqrt{2}, \dots, a_n\sqrt{n})$ and $v = (b_1, b_2/\sqrt{2}, \dots, b_n/\sqrt{n})$. $\square$

P3. The claim is **False**.

There is no inner product on $\mathbb{R}^2$ associated with the 1-norm, defined as $\|(x_1, x_2)\|_1 = |x_1| + |x_2|$. The 1-norm does not satisfy the Parallelogram Law (Theorem 9.3.6), and according to the text (p. 120), a norm can be used to define an inner product if and only if the norm satisfies the Parallelogram Law. As an example, let $u = (1, 0)$ and $v = (0, 1)$. Then

$$\|u + v\|^2 + \|u - v\|^2 = 2^2 + 2^2 = 8 \neq 4 = 2(1^2 + 1^2) = 2(\|u\|^2 + \|v\|^2).$$

P4. We prove a polarization identity for a real inner product space V .

For $u, v \in V$,

$$\langle u, v \rangle = \frac{\|u + v\|^2 - \|u - v\|^2}{4}.$$

Proof. Note that since V is a vector space over $\mathbb{R}$, we have $\langle v, u \rangle = \langle u, v \rangle$. Then

$$\begin{aligned}\|u + v\|^2 + \|u - v\|^2 &= \langle u + v, u + v \rangle - \langle u - v, u - v \rangle \\ &= (\|u\|^2 + \langle u, v \rangle + \langle v, u \rangle + \|v\|^2) - (\|u\|^2 - \langle u, v \rangle - \langle v, u \rangle + \|v\|^2) \\ &= 4 \langle u, v \rangle\end{aligned}$$

□